Semiconductor Manufacturing International Corporation

Doc. No.: TD-LO40-DC-2001	Doc. Title: 40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic Calibre Rule Based Dummy Fill	Doc.Rev: 3	Tech Dev. Rev: 1.14_0

Document Level: (For Engineering & Quality Document/工程暨品质文件专用)

☐ Level 1 - Manual ☑ Level 2 – Procedure/SPEC/Report ☐ Level 3 - Operation Instruction

Security Level:

☑ Security 1 - SMIC Confidential ☐ Security 2 - SMIC Restricted ☐ Security 3 - SMIC Internal

Document Change History

Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
2	1.13_0	2013-01-11	Hu Tang	Old Tech Version: 1.0 This document updates below items to make descriptions and using method clearer:A. Add “Vn dummy patterns can be inserted only when the switches "Mn_insert", "Mn+1_insert" and "Vn_inserted" are set to "YES", n = 1~7” in section 9 to notice the using method about via dummy insertion.B. Attached pdf file has been deleted because all contents are included by the main file in section 11C. Update file name in section 11 because attached package has been substituted by the runset file “SmicDR1R_calDMF40_log_sali_p1mtx_v1.0_2R.dmf”, and the content has been changed, and runset in attachment has been enhanced in technical, change items are as below:1. Enhance the NW and DNW constraints for AA dummy insertion2. Change runset format from SVRF into TVF3. Change the via dummy cell namePlease be kindly noted that the contents of this document may contain the confidential information of a third party such as SMIC's technology licensor, IP vendor, or customer, etc. Pursuant to the agreement between such third party and SMIC, SMIC's employee with a need to access such third party's confidential information shall (i) use such information solely for the benefit of SMIC or the third party, and for no other purpose, (ii) keep such information confidential for a period of fifteen years or longer from its disclosure of such information to SMIC, and (iii) agree that the third party is a third party beneficiary for the confidentiality obligation stated herein.
3	1.14_0	2017-06-13	Monica Cao	1. Update document title to “40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic Calibre Rule Based Dummy Fill”, Doc.Rev to “3R”, Tech Dev Rev to “V1.14_0” 2. 5.Reference,Update detail reference files, add reference design rule TD-LO40-DR-2001, update reference dummy insertion rule TD-LO40-DR-2009. 3. 6. Update Responsibility to “Technology R&D Center \ Tech-Bas Des-Enb Division \” 4. 11.Attachment: update attachment files, update dummy runset name to “SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.dmf”,and it enhance DUMBA outside AA insert. New add release file “SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.pdf” 5. Correct format in item 4 Nomenclature.

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1 Title

40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic Calibre Rule Based Dummy Fill

2 Purpose

This document is intended to be a release note for the users who adopt SMIC 40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic AA/Poly/Metal/Via layers Dummy Insertion Rule's runset for their design reference.

3 Scope

This document describes the Calibre dummy fill script of 40nm LL release. Switches selection will be discussed within.

4.Nomenclature: NA

5 Reference

5.1 reference design rule

40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V Low Leakage And Generic Design Rule

Doc.No.: TD-LO40-DR-2001

Doc.Rev.: 6R

Tech Dev Rev:1.12

5.2 reference dummy insertion rule

40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic AA/Poly/Metal/Via layers Dummy Insertion Rule

Doc.No.: TD-LO40-DR-2009

Doc.Rev.: 5R

Tech Dev Rev:1.14

6 Responsibility

Technology R&D Center \ Tech-Bas Des-Enb Division \

7 Calibre dummy insertion script options setup

This script will select to fill dummy patterns for AA, GT, M1~M8, V1~V7, TM1, TM2, MTT2, ALPA layers according to user's setting. The switches in the script are as below:

#DEFINE AA_insert YES

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#DEFINE GT_insert YES

#DEFINE M1_insert YES

#DEFINE V1_insert YES

#DEFINE M2_insert YES

#DEFINE V2_insert YES

#DEFINE M3_insert YES

#DEFINE V3_insert YES

#DEFINE M4_insert YES

#DEFINE V4_insert YES

#DEFINE M5_insert YES

#DEFINE V5_insert YES

#DEFINE M6_insert YES

#DEFINE V6_insert YES

#DEFINE M7_insert YES

#DEFINE V7_insert YES

#DEFINE M8_insert YES

#DEFINE TM1_insert YES

#DEFINE TM2_insert YES

#DEFINE MTT2_insert NO

#DEFINE ALPA_insert YES

Set the switch to "YES", dummy insertion script will fill the dummy pattern, otherwise set to "NO". Please only set one switch to "Yes" between "TM2_insert" AND "MTT2_insert", while you must keep the other one to "NO".

8 Example for switches setup

Here will show you an example on how to set the switches for dummy insertion. Suppose that you only need to fill AA dummy and GT dummy patterns, then the switches should be set as below:

#DEFINE AA_insert YES

#DEFINE GT_insert YES

#DEFINE M1_insert NO

#DEFINE V1_insert NO

#DEFINE M2_insert NO

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#DEFINE V2_insert NO

#DEFINE M3_insert NO

#DEFINE V3_insert NO

#DEFINE M4_insert NO

#DEFINE V4_insert NO

#DEFINE M5_insert NO

#DEFINE V5_insert NO

#DEFINE M6_insert NO

#DEFINE V6_insert NO

#DEFINE M7_insert NO

#DEFINE V7_insert NO

#DEFINE M8_insert NO

#DEFINE TM1_insert NO

#DEFINE TM2_insert NO

#DEFINE MTT2_insert NO

#DEFINE ALPA_insert NO

9 Important notice

This script will insert two types metal dummy (MnDUM and MnDOP) for Mn, n = 1~8.

"BORDER" layer must be drawn as chip edge in your layout in order to insert dummy patterns in it. This dummy insertion script must be run with hierarchy mode.

Vn dummy patterns can be inserted only when the switches "Mn_insert", "Mn+1_insert" and "Vn_inserted" are set to "YES", n = 1~7.

10 Known Issue

No known issue.

11 Attachment

SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.dmf

SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.pdf

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Change History:

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Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
0R	1.0	2012-08-23	Hu Tang	Initiate
1R	1.0	2012-09-27	Hu Tang	Enhance hierarchy structure to reduce gds result size for AA, GT, metal dummy patterns
2R	1.0	2013-01-11	Hu Tang	This document updates below items to make descriptions and using method clearer: A. Add “Vn dummy patterns can be inserted only when the switches "Mn_insert", "Mn+1_insert" and "Vn_inserted" are set to "YES", n = 1~7” in section 9 to notice the using method about via dummy insertion. B. Attached pdf file has been deleted because all contents are included by the main file in section 11 C. Update file name in section 11 because attached package has been substituted by the runset file “SmicDR1R_calDMF40_log_sali_p1mtx_v1.0_2R.dmf”, and the content has been changed, and runset in attachment has been enhanced in technical, change items are as below: 1. Enhance the NW and DNW constraints for AA dummy insertion 2. Change runset format from SVRF into TVF 3. Change the via dummy cell name
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